

HARBORLINE DESIGN DOCUMENTATION

This note records how Harborline answers employee questions: a hand-written orchestrator, one MCP server, and OpenAI embeddings with an in-process cosine index. The two People Desk demos follow fixed tool sequences.

DESIGN CHOICES

Orchestration. `harborline/agent.py` classifies intent with regular expressions and then runs a fixed workflow. The model does not choose tools. That keeps HR paths repeatable: the same question always calls the same tools, and the trace lists each name, argument, and result. When `HARBORLINE_ANSWER_MODE=llm` and `OPENAI_API_KEY` is set, the chat model only rewrites the draft after the tools return. Refusals, missing records, and thin evidence stay on that draft.

MCP server. One FastMCP server, named `harborline`, exposes eight tools. Policy tools read the RAG index. Profile, PTO, and benefits tools read mock HarborHub JSON under `data/`. Ticket and email tools are session-only mocks and do not write `data/tickets.json`. The agent discovers tools with `tools/list` and calls only names from that list through `tools/call`. FastMCP builds each JSON Schema from the Python signature, so the schema stays tied to the function.

Tool schemas.

- `search_policy_documents`: query (required), `employee_id`, kind
- `get_policy_section`: `policy_id` (required), section
- `lookup_employee_profile`: `employee_id`
- `check_pto_balance`: `employee_id`
- `lookup_benefits_status`: `employee_id`
- `create_mock_hr_ticket`: topic, summary, `employee_id`, confirm
- `draft_hr_email`: `employee_id`, subject, body, confirm
- `check_policy_compliance`: scenario, `employee_id`, `policy_id`

Transport. The CLI and Cursor use stdio: the client spawns `python -m harborline.mcp_server` and speaks JSON-RPC on stdin and stdout, with no open port. The People Desk page, `POST /chat`, and `POST /demos/{id}` use `mcp-inproc`, the same FastMCP object inside the web process, so a Render container does not spawn a second Python process per request. Streamable HTTP on `127.0.0.1:8765` is available when an external client needs an HTTP MCP endpoint.

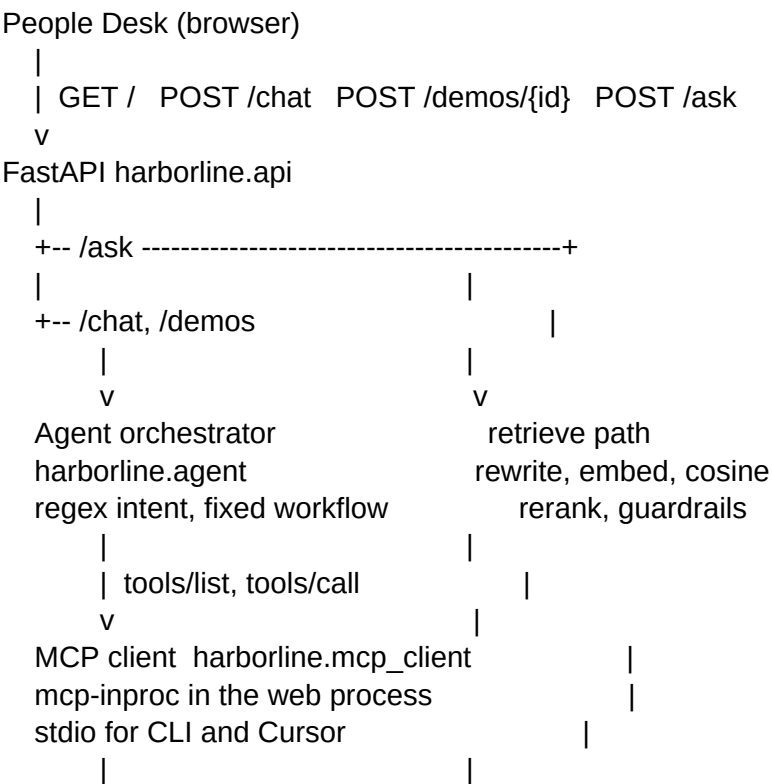
Embeddings and chunking. Embeddings are OpenAI `text-embedding-3-small` (1536 dimensions) from the embeddings API. Ingest splits each handbook by heading. A section longer than 900 characters is cut into windows of 900 with 120 characters of overlap, so a policy fact stays with its heading and neighboring sentences still share context.

Retrieval k and index. Search embeds the rewritten query, takes a candidate pool of HARBORLINE_FETCH_K=20, then reranks to HARBORLINE_TOP_K=5. The blend is 0.65 cosine and 0.35 lexical overlap, and the reranker prefers distinct source files. The score floor is HARBORLINE_MIN_SCORE=0.22. Vectors are L2-normalized and cached under .cache/. Query time ranks that matrix in the process. tfidf remains the offline path used by pytest and CI.

Deployment. A Docker image runs FastAPI with uvicorn on port 8000. The image copies corpus/, data/, and eval/ and does not download local embedding weights. On Render, set OPENAI_API_KEY and HARBORLINE_ANSWER_MODE=llm. The Dockerfile pins retrieve, so the service variable overrides it. The first search after a restart embeds the corpus and holds the matrix in that process. The cache does not survive a new deploy.

Safety guardrails. Out-of-corpus questions (stock tips, other employers, medical advice, and similar patterns) are refused. Empty retrieval or a top score under 0.22 returns a weak-match refusal. Missing employee ids ask for clarification. Missing HarborHub rows escalate. Mock ticket and email writes stay pending until confirm=true, and even then they exist only in that process. The model rewrite leaves refusals in place for an unavailable tool bus, a missing employee, or incomplete policy evidence.

ARCHITECTURE



```

      v
MCP server harborline.mcp_server (FastMCP)
|
+-- search_policy_documents
|  get_policy_section
|  check_policy_compliance ----->
|    RAG index
|    OpenAI embeddings + cosine cache
|    corpus/ policies
|
+-- lookup_employee_profile
|  check_pto_balance
|  lookup_benefits_status
|    mock HarborHub JSON in data/
|
+-- create_mock_hr_ticket
|  draft_hr_email
|    session-only MOCK
|
After tools, if HARBORLINE_ANSWER_MODE=llm:
      v
OpenAI chat API
gpt-4o-mini, temperature 0, seed 42
writes the answer from the draft, snippets, trace

```

DEMO TASKS

Both demos are fixed in harborline/demos.py. POST /demos/{id} and POST /chat with the same question and employee id run the same workflow. The client opens the MCP bus, lists tools, then makes the calls below.

Demo 1. Remote eligibility (remote-emp-1008).

Question: Am I eligible for fully remote work living in Tacoma?

Employee: EMP-1008 (Alex Kim). Intent: remote_eligibility.

1. lookup_employee_profile
employee_id=EMP-1008
Reads the HarborHub profile, including location category and miles from the hub.
2. search_policy_documents
query=remote hybrid hub 50 miles office days POL-RMT-003
kind=policy
Retrieves hub-rule passages.
3. get_policy_section
policy_id=POL-RMT-003
section=Location categories

Loads that heading for the citation.

4. check_policy_compliance
scenario=the original question
employee_id=EMP-1008
policy_id=POL-RMT-003
Returns a verdict.

The draft states that Alex Kim is hub-coded, about 32 miles from Seattle, and that fully remote work needs a People Operations reclass. The trace escalates for that reclass. Citations come from corpus/03-remote-hybrid-work.md.

Demo 2. Benefits election (benefits-emp-1008).

Question: What medical plan and 401k deferral do I have?

Employee: EMP-1008. Intent: benefits.

1. search_policy_documents
query=benefits 401k medical enrollment POL-BEN-006
kind=policy
2. get_policy_section
policy_id=POL-BEN-006
section=US medical
3. get_policy_section
policy_id=POL-BEN-006
section=Retirement
4. lookup_benefits_status
employee_id=EMP-1008
Reads the mock election: HDHP medical plan and a 3 percent 401(k) deferral.

The draft quotes the 100 percent match on the first 4 percent deferred, immediate vesting, and the HarborHub election row. Citations come from corpus/06-employee-benefits.md.

With HARBORLINE_ANSWER_MODE=llm, both demos then send that draft, the citations, and the tool summaries to the OpenAI chat API. The tool order above stays the same.